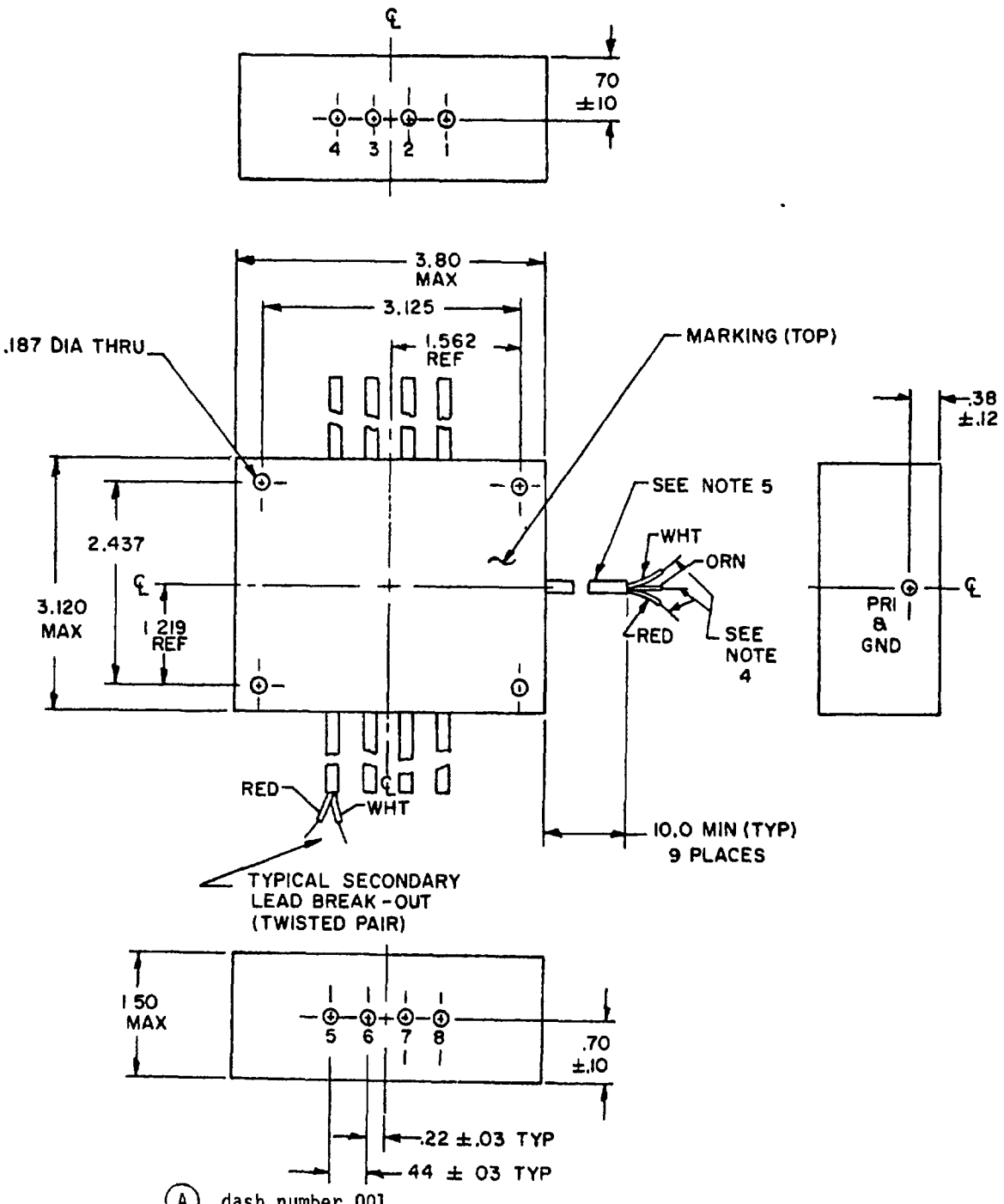
MILITARY SPECIFICATION SHEET

TRANSFORMER, PULSE, LOW POWER, TYPE TP5SXNNNNBZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the transformer described herein shall consist of this specification and the latest issue of MIL-T-21038.



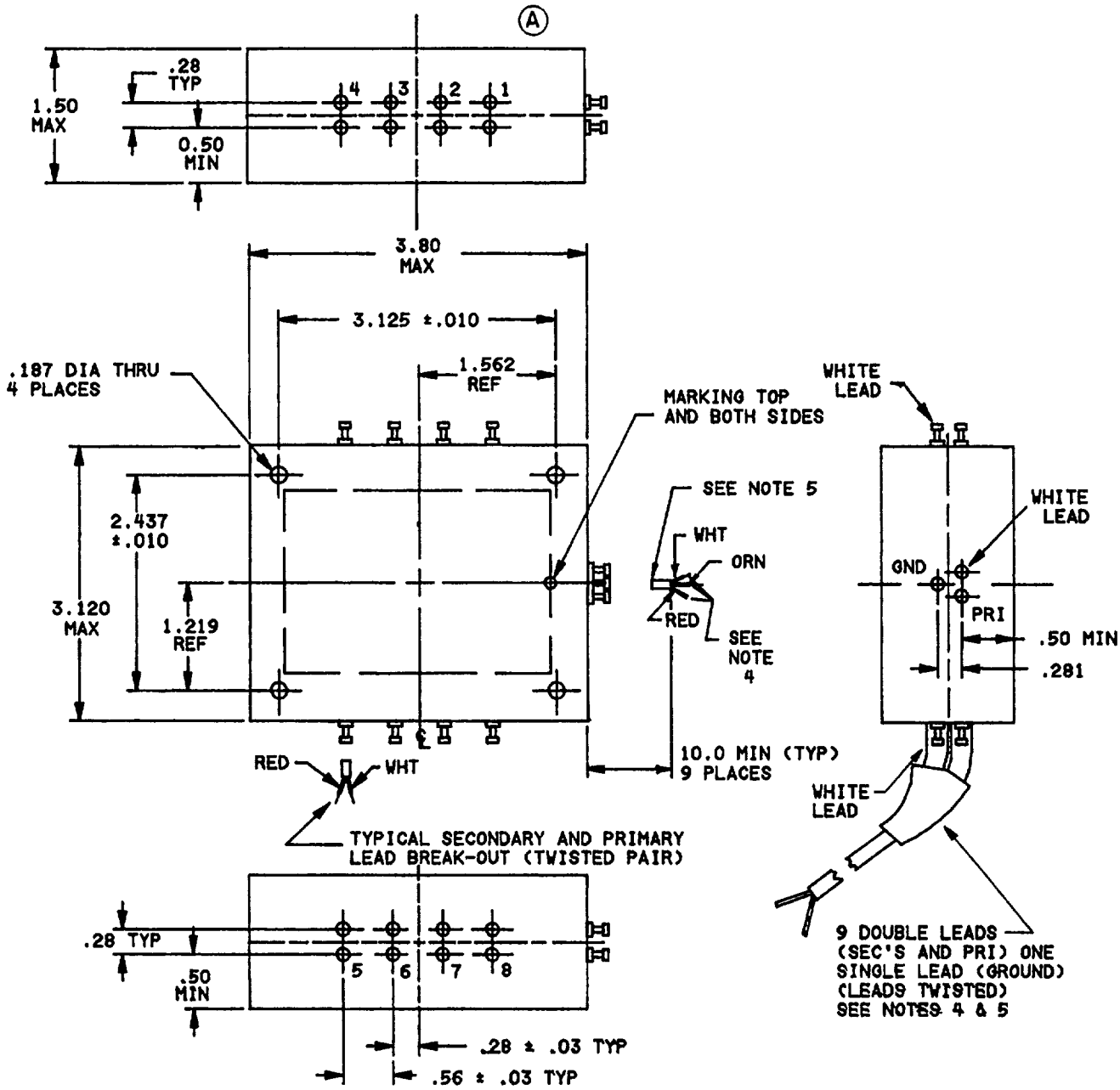
(A) dash number 001

FIGURE 1. Dimensions and configuration.

(A) denotes changes

FSC 5950

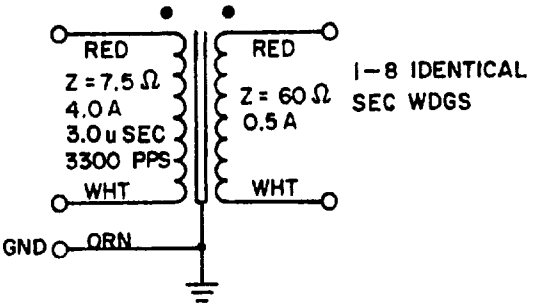
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(A) dash number 002

FIGURE 1. Dimensions and configuration - Continued.

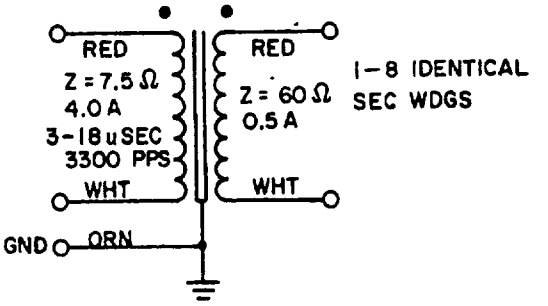
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50 WV(PK) 5500
(WV(PK) 1500 SEC TO SEC)

CIRCUIT DIAGRAM (dash 001)

Ⓐ



50 PWV 5500
(PWV 1500 SEC TO SEC)

CIRCUIT DIAGRAM (dash 002)

INCHES	MM	INCHES	MM
.03	0.8	.50	12.7
.10	2.5	.56	14.2
.010	0.25	.70	17.8
.12	3.0	1.219	30.96
.187	4.75	1.50	38.1
.22	5.59	1.562	39.67
.28	7.1	2.437	61.90
.281	7.14	3.120	79.25
.38	9.7	3.125	79.38
.44	11.18	3.80	96.5
		10.0	254.0

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on all dimensions shall be ±.01 (0.3 mm) unless otherwise specified.
4. All leads shall be #24 AWG per MIL-W-16878/4, type E.
5. Each twisted terminal group shall be covered with clear, #11 AWG polytetrafluoroethylene sleeving per MIL-I-22129.

FIGURE 1. Dimensions and configuration - Continued.

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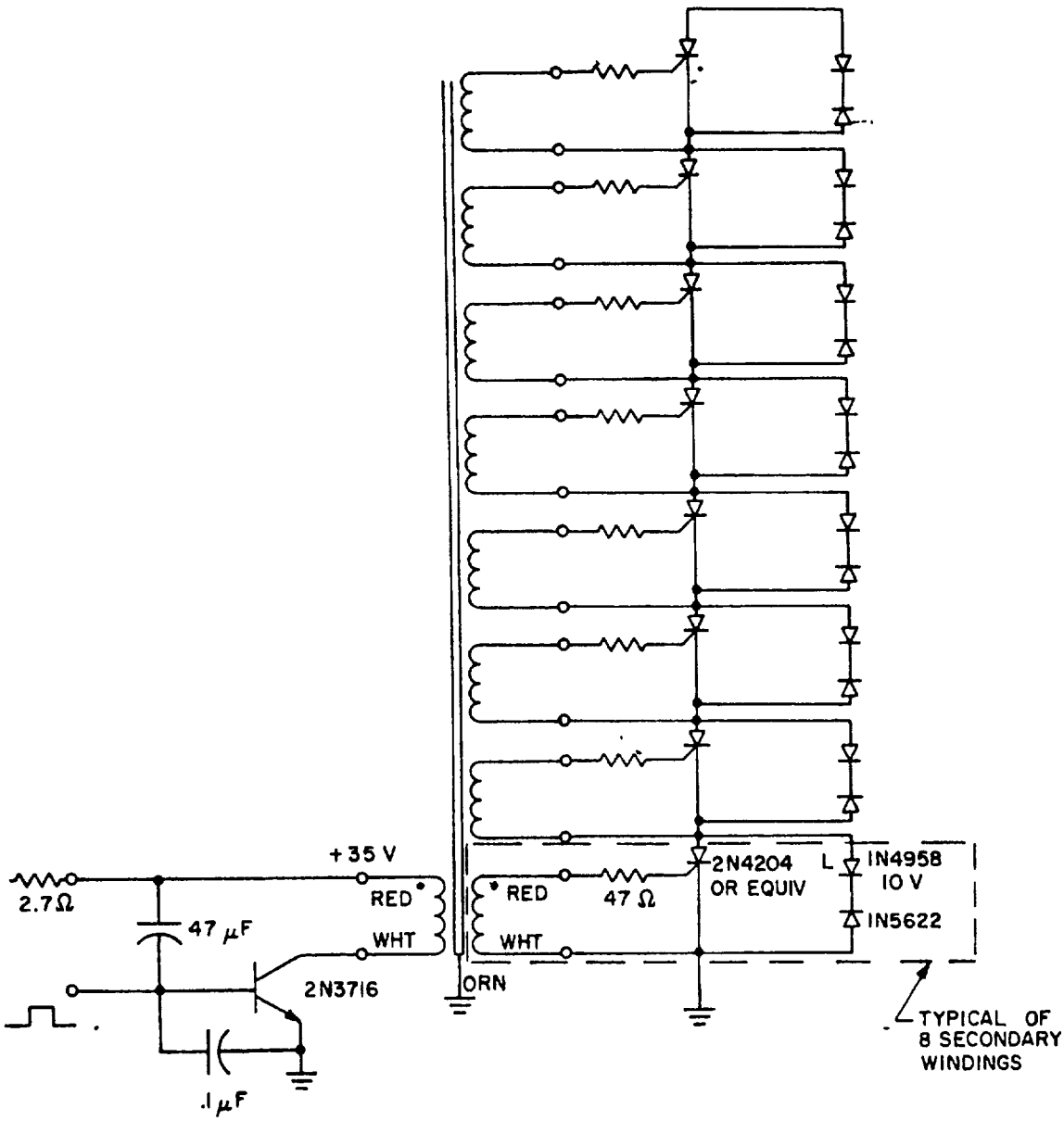
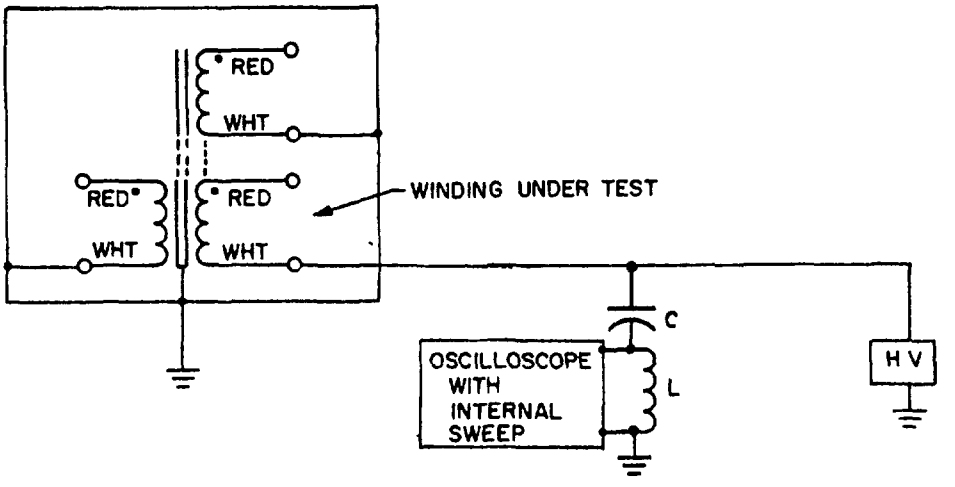


FIGURE 2. Application modulation switch test circuit.

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NOTES:
1. When using circuit, ground the case of the transformer, and all windings except that being tested.
2. Legend for test circuits: C = 200 picofarad mica capacitor, corona free; L= RF choke, 20 to 30 millihenries inclusive, with a minimum Q of 50 at 100 kilohertz; HV = high voltage dc source, corona free.
3. Corona will be evident as a superimposed high-frequency oscillation on the basic power wave.
(A) 4. Test 1; as shown voltage applied between each secondary.
(A) 5. Test 2; voltage applied between all secondaries tied together to primary and ground.

FIGURE 3. Corona test circuit.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Turns ratio: 1:1:1:1:1:1:1:1:1 ±3 percent.

Ⓐ Duty cycle: 0.01 at 3 µs, 0.06 at 18 µs.

Ⓐ Average power rating: 1.2 watts at 3 µs, 7.2 watts at 18 µs.

Case: Encapsulated

Ⓐ Terminals: Turret terminals with attached insulated wire leads per MIL-W-16878/4, type E.

Weight: 2.0 pounds maximum.

Terminal strength: MIL-STD-202, method 211 , test condition A - pull 2.5 pounds.

Dielectric withstanding voltage:

At atmospheric pressure: MIL-STD-202, method 301, 8,700 volts rms between secondary and ground with all secondaries tied together; 3,100 volts rms between secondary and secondary; and 100 volts rms between primary to secondary and ground.

At reduced barometric pressure: Not applicable.

Insulation resistance: MIL-STD-202, method 302, test condition B. Not less than 10,000 megohms.

Electrical characteristics: See table I.

Voltage-time product (ET constant): The voltage-time product shall be determined by the amplitude of the applied pulse by the time required for the magnetizing current to deviate 20 percent from the value of the linearly extrapolated ramp portion of the magnetizing current wave form in the region of core saturation.

Ⓐ No load excitation current: 0.9 A max. pK at 30 volts, 18 µs applied to primary.

Ⓐ Secondary capacitance: The direct capacitance of each shorted secondary winding to all other windings and ground shall be 18 picofarads ± 20%.

Test circuit: See figure 2.

Ⓐ Corona: Test number 1; a test voltage of 1750 volts dc is applied between each secondary winding. Test number 2; a test voltage of 7150 volts dc is applied between all secondary windings tied together to the primary and ground. The corona shall not exceed 1 inch peak-to-peak deflection as indicated on the oscilloscope using the test circuit specified (see figure 3). The test voltage shall be applied at atmospheric pressure in the manner specified for the dielectric withstanding voltage test.

Life: See MIL-T-21038, life expectancy X.

Ambient temperature: 71°C.

Temperature rise: 59°C (1.2 W).

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Shock (specified pulse): MIL-STD-202, method 213, test condition H.

Salt spray: MIL-STD-202, method 101, test condition B.

a. Salt solution concentration: 5 percent.

b. Examination after exposure: The transformer shall be thoroughly washed. The temperature shall not exceed 38°C. The transformer shall be placed in an oven maintained at 50°C ± 3°C for a period of 24 ± 4 hours. At the end of this period, the transformer shall be removed from the oven and examined for corrosion. There shall be no evidence of corrosion as exhibited by any visible degradation of the surfaces than can be attributed to flaking, pitting, blistering, or otherwise loosened coating of metal surface.

Part number: M21038/25-(dash number from table I)

TABLE I. Electrical characteristics including waveform parameters.

Dash number	Impedance (ohms max)		current (A pk) nom		Voltage (V pk)		DCR (ohms max)		Pulse duration (µs max)	Pulse rise time (µs max)	Droop (% max)	Pulse repetition frequency (Hz)	Primary ET constant (V µs min)	Inductance (µH min) 2/
	Source	Load 1/	Pri	Sec 1/	Pri	Sec 1/	Pri	Sec 1/						Pri
001	7.5	60	4.0	0.5	30	30	0.144	0.75	3.0 at 70%	0.15	10	3,300	500	0.25
Ⓐ 002	7.5	60	4.0	0.5	30	30	0.144	0.75	3.0-18.0 at 70%	0.40	10	3,300	500	600

1/ Across each secondary winding.
2/ Inductance is measured at 1 volt, 1 KHz.

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - SH, AS, OS
Air Force - 17, 99
DLA - ES

User activities:
Army - ME
Navy - MC
Air Force - 19

Agent:
DLA-ES

Preparing activity:
Navy - EC

(Project 5950-0634)

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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

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c. Reason/Rationale for Recommendation

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